Proof by Induction-Series

Key Points

Remember to prove a statement holds for all positive integers by induction, we do the following:

- Show it holds for the base case 1
- Assume it holds for a case k
- Prove the statement then holds for the case $k+1$

For a sequence $(a_n)_{n=1}^{\infty}$, identities of the form $s_N = \sum_{i=1}^{N} a_i = f(N)$ can be proven by induction. For the inductive step assume $s_k = f(k)$, and show $s_{k+1} = s_k + a_{k+1} = f(k+1)$

Basic Skills

Q1. Given that $a_1 = 1, a_2 = 2, a_3 = 2, a_4 = 1$, calculate

a) s_1 b) s_2 c) s_3 d) s_4

Q2. Given a sequence given by $a_n = n$, calculate

a) s_1 b) s_2 c) s_3 d) s_5 e) s_{10}

Q3. Consider $f(x) = \frac{x}{2x+1}$. Given that $a_1 = \frac{1}{3}, a_2 = \frac{1}{15}$ and $a_3 = \frac{1}{35}$

a) Verify that $s_1 = f(1),\ s_2 = f(2), s_3 = f(3)$

b) It is now belived that $s_n = f(n)$ for all $n \geq 1$, state the base case from part a)

c) State what you would have to show to prove that $s_n = f(n)$ for all $n \geq 1$ (Note. You do not have to show it)

Competency Skills

Q4. Consider the sequence $a_n = n$. We want to show that the formula $s_N = \sum_{i=1}^{N} a_i = \frac{1}{2}N(N+1)$ holds using proof by induction

a) Verify the base case

b) Assuming that $s_k = \frac{1}{2}k(k+1)$, show that $s_{k+1} = \frac{1}{2}(k+1)(k+2)$

c) Write out the statement we can conclude from parts a) and b) using proof by induction

Q5. Consider the sequence $a_n = n^2$. Show the formula $s_N = \sum_{i=1}^{N} a_i = \frac{1}{6}N(N+1)(2N+1)$ using a proof by induction

Q6. Verify the formula $\sum_{i=1}^{N} \frac{1}{i(i+1)} = \frac{N}{N+1}$ using proof by induction

Q8. Verify the formula for the sum of an infinite geometric series where the first term is a, and the common ratio is given by $r < 1$ by using proof by induction

Advanced Skills

Q8. Prove the product formula $\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{N^2}\right) = \frac{N+1}{2N}$ for $N \geq 2$ by induction

Q9. Verify that $1^3 + 2^3 + 3^3 + \cdots n^3 = (1 + 2 + 3 + \cdots n)^2$. (Can you use Q4 to help?)

Q10. Find and verify a formula for the sum of the first N odd numbers by induction

Q11. Verify that $1 + 2x + 3x^2 + \cdots + n\,x^{n-1} = \frac{1-(n+1)x^n + nx^{n+1}}{(1-x)^2}$, provided $x \neq 1$.

Extension

Q12. Show that $\sum_{i=1}^{N} \frac{1}{i^2} < 2$, for any positive integer N

Hint: Actually try to show the stronger statement that $\sum_{i=1}^{N} \frac{1}{i^2} < 2 - \frac{1}{N}$